# SME Credit Default: Machine Learning Comparison

## Student files

- `assignment_sme_credit_risk_en.docx`: the assignment brief.
- `sme_credit_default_1000x20_en.csv`: the data, for R/Python and other statistical tools.
- `sme_credit_default_1000x20_en.xlsx`: equivalent data plus variable and category dictionaries.

## About the dataset

This is an entirely synthetic dataset of 1,000 small and medium-sized enterprises (SMEs), each with 20 predictors and a binary response (`default_12m`). A value of 1 indicates a default within 12 months and 0 indicates no default within that period. The dataset contains 284 positive cases and no missing values. Variable names and categorical values are in English.

The teaching exercise intentionally does not disclose the data-generating mechanism or precomputed model scores. All preprocessing decisions and interpretations should be based on the supplied data and the training folds only.

The dataset is a methodological teaching artifact, not a validated model of real credit risk.

## Import into R

```r
df <- read.csv("sme_credit_default_1000x20_en.csv", stringsAsFactors = TRUE)
str(df)
table(df$default_12m)
```

## Import into Python

```python
import pandas as pd
df = pd.read_csv("sme_credit_default_1000x20_en.csv")
print(df.shape)
print(df["default_12m"].value_counts())
```